MILITARY SPECIFICATION

CARTRIDGE, REFERENCE, CALIBER .38, SPECIAL, (M41 TYPE)

1. SCOPE

1.1 This specification covers a cartridge to be used for calibration purposes in Caliber .38, Special, Ball, M41 proof testing equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-45208 — Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-636 —Visual Inspection Standards for Small Arms Ammunition Through Caliber .50

DRAWINGS

MUNITIONS COMMAND

C8595425 — Cartridge, Reference, Caliber .38, Special, (M41 Type)

C7643674 — Classification of Cartridge Case Defects

F10524420 — Packing and Marking, Cartridges, Reference, Caliber .38, Special, (M41 TYPE); Cartons, Box, Ammunition, M2A1; Box, Wirebound

IEL-8595425 — Inspection Equipment List for Cartridges, Reference, Caliber .38, Special, (M41 Type)

PUBLICATIONS

MUNITIONS COMMAND

AMCR 700-700-Vol. III Manual of Test Methods for Small Arms Ammunition

AMCR 715-505– Ammunition Ballistic Acceptance Test Methods — Test Procedures for 7.62 mm Cartridges

FSC 1305

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ORD-SIP-S324–Standard Inspection Procedures for Packing of Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The cartridge shall comply with Drawing C8595425, referenced specifications and the following:

3.2 Bullet extraction. The force required to extract the bullet from the cartridge case shall be not less than 50 pounds.

3.3 Residual stress. The cartridge case shall be free of residual stress.

3.4 Waterproof. The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 45 seconds.

3.5 Velocity. The grand average of the velocities of all the sample cartridges, conditioned at 70° ± 2° Fahrenheit (F), shall be 950 feet per second (ft/see), plus or minus 20 ft/sec, at 15 feet from the muzzle of the weapon. The average velocity of each sample of 20 cartridges shall not vary from the grand average by more than plus or minus 10 ft/sec. The standard deviation of velocities for each sample of 20 cartridges shall not exceed 20 ft/sec.

3.6 Chamber pressure. The average chamber pressure of the sample cartridges, conditioned at 70° ± 2° F, shall not exceed 16,000 psi.

3.7 Function and casualty. The cartridge shall function without casualty.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.8.2 Foreign matter. The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection. Inspection shall be in accordance with Specification MIL-I-45208, except as specified herein.

4.2 Inspection provisions. Use of alternative inspection procedures and inspection equipment shall be in accordance with Specification MIL-I-45208, except that in all ballistic testing, the use of the equipment specified and the methods of test prescribed in this specification and other documents referenced herein shall be mandatory.

4.2.1 Lot.

4.2.1.1 Presentation of lots. The product shall be presented in accordance with Standard MIL-STD-105. The cartridge lot shall be restricted to cartridges assembled with cases primed with primers from one lot only, using an established charge of propellant

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from one lot only. The entire lot of reference cartridges shall be assembled and charged on a single loading machine (see 6.2).

4.2.1.2 Size of lot. The cartridge lot size shall be as agreed upon between the supplier and the contracting officer.

4.2.1.3 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity.

4.2.2 Examination. Examination shall be performed as indicated below:

4.2.2.1 Major and minor defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, table I, using applicable sampling plans and acceptance criteria of Standard MIL-STD-105.

4.2.2.2 Acceptable quality levels. The acceptable quality levels for visual and dimensional defects shall be as follows:

Defect class	Percent
Major	0.25
Minor	1.50

4.2.2.3 Classification of defects. The visual and dimensional defects shall be classified as specified in table I.

TABLE I

No.	Defect and method of inspection	Major	Minor	Major or minor
	Visual [1]			
	Cartridge			
1	Discolored, dirty, oily, smeared		X	
2	Corroded or stained, if etched	X		
3	Mixed ammunition types	X		
	Case			
4	Round head	X		
5	Dent			X
6	Split case	X		
7	Perforated case	X		
8	Draw scratch			X
9	Scratch		X	
11	Scaly metal			X
12	No chamfer on head (rim)	X		
13	Wrinkle		X	
14	Bulge		X	
15	Illegible or missing head stamp		X	
16	Defective head		X	
17	Defective mouth		X	
	Bullet			
18	Dent		X	
19	Scratch		X	
20	Split bullet jacket	X		
21	Loose bullet	X		
22	Scaly metal			X
23	Upset (crooked) point		X	
24	Exposed steel (clad jacket)		X	
25	Flat point		X	
	Primer			
26	No primer	X		
27	Cocked primer	X		
28	Inverted primer	X		
29	Loose primer	X		
30	Nicked or dented primer		X	
31	No waterproofing material (primer pocket joint)		X	

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TABLE I—Continued

No.	Defect and method of inspection	Major	Minor	Major or minor
	Gaging			
32	Total length	X		
33	Cartridge profile failure (requiring) more than 5 lbs dead weight to insert in profile and alignment gage)	X		
34	Diameter of head	X		
35	Thickness of head	X		
36	Depth of primer	X		

[1] Refer to Standard MIL-STD-636 (caliber .45 section) for visual defect standards for defects 1 through 31. In the event of conflict between table I of this specification and Standard MIL-STD-636 as to defect classification, the classification specified in table I shall apply.

4.2.3 Tests. The tests listed in table II shall be conducted in accordance with the methods and procedures specified in 4.3.

4.2.3.1 Test samples. The quantities for the various tests shall be as specified in table II. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot, The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

Test	Number of cartridge	Requirement paragraph
Bullet extraction [1]	25	3.2
Mercurous nitrate (Residual stress) [2]	20	3.3
Waterproof [3]	50	34.
Velocity [4]	300	3.5
Chamber pressure [4]	300	3.6
Function & casualty [5]	.....	3.7

[1] Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample. the lot shall be rejected.

[2] The occurrence of splits in two or more cartridges shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges arc found in the second sample, the lot shall be rejected.

[3] Failure of ten or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three, but less than ten, cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. The lot shall be rejected if, in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirement.

[4] Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot (see 6.2.1).

[5] Examination for firing defects shall be made during the velocity and chamber pressure tests. A lot shall be rejected when firing defects exceed the acceptance number in table III.

4.2.3.2 Firing defects. Firing defects and acceptance numbers shall be as specified in table III.

TABLE III

Defects	Acceptance number
1. Misfire	1
2. Bullet remaining in bore	0
3. Primer leak	
(a) Perforation in firing pin indent in primer cup	9
(b) Escape of gas through primer cup other than 3a	3
(c) Escape of gas around primer cup [1]	12
(d) Blown primer or primer falls out of pocket on retraction of bolt	1
(e) Primer remains in pocket, but is physically loose	3
4. Case casualties	
(a) Longitudinal split [2]	

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TABLE III—Continued

Defects	Acceptance number
(1) Mouth (I)	12
(2) Body (J)	9
(3) Body (K)	2
(4) To head (L)	1
(5) Through head (M)	1
(b) Circumferential rupture[2]	
(1) Partial (J)	3
(2) Partial, body (K)	1
(3) Partial, head (L)	1
(4) Complete	1

[1] Gas escapes around more than 50% of periphery of cup.
[2] For location of defects indicated by letters in parenthese, see Drawing C7643674.

4.2.4 Packaging, packing and marking inspection. Inspection for packaging, packing and marking shall be in accordance with using the following AQL'S:

- Publication ORD-SIP-S324 as applicable to the drawing (or as required by the contract)

	AQL Major percent	AQL Minor percent
(a) Contents of container		
Type of packs		
Carton packs	1.0	2.5
(b) Packed and sealed containers		
Gasket sealed container		
Physical characteristics, paint marking	1.0	2.5
Gasket sealing	1.0	...
(c) Overpacks and overpack contents		
Wirebound boxes (crates)	1.5	6.5

4.2.5 Inspection equipment. The examination and tests shall be made using the equipment listed on IEL-8575425, except as specified in 4.2.

4.3 Test methods and procedures.

4.3.1 Bullet extraction. The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three, nor more than six, inches per minute.

4.3.2 Mercurous nitrate. The test shall be conducted in accordance with Publication AMCR 715-505 (for 7.62 mm cartridge).

4.3.3 Waterproof. The test shall be conducted in accordance with Publication AMCR 715-505 (for 7.62 mm cartridges). The container shall be evacuated to a pressure of 5 pounds per square inch below atmospheric pressure and held at that pressure for 45 seconds.

4.3.4 Velocity. The test shall be conducted in accordance with the general procedures of Publication AMCR 700–700 Vol. III, except that instrumental velocity shall be taken at 15 feet from the muzzle of the test weapon. The distance between screens shall be 20 feet, and the cartridges shall be fired in barrels previously fired 100 to 300 times. Twenty cartridges shall be fired in each of five velocity barrel assemblies. The average velocity and standard deviations of velocities for each velocity barrel assembly and the average velocity for the five assemblies shall be computed. If the average velocity for any velocity barrel assembly varies from the average for the five assemblies by more than plus or minus 10 ft/sec, or if the standard

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deviation of velocities for any assembly exceeds 20 ft/sec, that assembly shall be discarded and another selected and firing conducted therein. An additional twenty cartridges shall be fired in the same manner in the same velocity barrel assemblies on each of two succeeding days. The average velocity and standard deviation of velocities for each 20-round test, the daily average velocity (20 rounds in each of 5 assemblies) and the grand average for the entire 15 velocity series (5 assemblies on each of 3 days) shall be computed.

4.3.5 Chamber pressure. The tests shall be conducted in accordance with the general procedures of Publication AMCR 700-700 Vol. III and the following: The cartridges shall be fired in barrel's previously fired 100 to 300 times. Twenty cartridges shall be fired in each of 5 pressure barrel assemblies on each of 3 days. Upon completion, the average presure for the entire 15 pressure series (5 assemblies on each of 3 days) shall be computed. Prior to the firing of each 20 round test, 2 shots shall be fired for seating purposes only. Wax filled gas checks shall be changed after firing each series of 2 seating shots and the 20 shots for record (22 shots).

4.3.6 Function and casualty. The test shall be conducted simultaneously with the velocity and chamber pressure tests.

4.3.7 Defect penalty. In any ballistic test in which the occurrence; of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with table III.

5. PREPARATION FOR DELIVERY

5.1 Packing. Level A (Worldwide shipment). The cartridges shall be packed in accordance with Drawing F10524420, or as required by contract,

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with the applicable Drawing F10524420 cited in 5.1.

6. NOTES

6.1 Ordering data. Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.2 Uniformity of propellant charge. A machine should be chosen which is known to drop charges with better than average uniformity and which gives constant results in the daily production tests. If relative machine performance is unknown, special tests should be fired to assure selection of a good machine.

6.2.1 A quantity of 20 rounds should be fired for velocity and 20 rounds for pressure each four hours of an eight hour shift to verify uniformity of charge weight and established velocity and pressure values.

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